

Equations

1. Carnot Efficiency

a. $\frac{T_h - T_c}{T_h}$ or $1 - \frac{T_c}{T_h}$

i. T_h = Temperature hot

ii. T_c = Temperature cold

b. Fundamental efficiency of heating up objects, always lower than 1

c. $\frac{T_c}{T_h - T_c}$ Is also a thing but we haven't used this once so --

2. Wind

a. $Momentum = m * v = pALv$

i. m = mass

ii. v = velocity

iii. p = density

iv. A = area of the blade circle

v. L = length of the air / water column

b. Plugs in nicely into KE equation $\frac{mv^2}{2} = \frac{pALv^2}{2}$ Joules

c. Force = $\frac{Momentum}{time} = \frac{pALv}{t} = pAv * \frac{L}{t} = pAv^2$ Newtons

i. $\frac{L}{t} = \frac{Distance}{time} = velocity$

d. Power = $\frac{KE}{time} = \frac{pALv^2}{2t} = \frac{pAv^2}{2} * \frac{L}{t} = \frac{pAv^3}{2}$ Watts

i. Should usually be multiplied by Power Coefficient C_p

e. As seen this is highly dependent on velocity

f. Torque = $r * (Flift - Fdrag)$

i. Power = torque * rotation speed

g. Capacity Factor = $\frac{Actual\ Production}{Theoretical\ 365/24\ generation}$

i. Onshore 0.1 - 0.4

ii. Offshore 0.2 - 0.6

3. Biomass

a. Gasifier Efficiency =
$$\frac{LHV_{gas} * m_{gas}}{LHV_{biomass} * m_{biomass}}$$

i. LHV = Lower Heating Value (product is water vapour)(given)

ii. m = mass

4. Hydropower

a. $PE = mgh$

b. $KE = \frac{mv^2}{2}$

i. g = gravitational acceleration ($9.81 \frac{m}{s^2}$)

ii. m = mass

iii. h = height

5. Thermodynamics (oh god)

a. Disclaimer

i. U, H, S, G = normal values

ii. u, h, s, g = molar value

b. Work

i. Work = Energy

1. $W = Fx$

a. F = Force

b. x = displacement

2. $P = \frac{F}{A}$

a. P = Pressure

b. A = Area

c. $F = PA$

i. $W = PAx$

1. $W = PV$

- a. $V = \text{Volume}$

3. $\text{Work} = \text{Pressure} * \text{Volume}$

- ii. $Q_{\text{hot}} = Q_{\text{cold}} + W_{\text{compression}}$

1. $Q = \text{heat}$

c. Internal Energy (U) & Enthalpy (H)

- i. Sum of all thermal energy of all atoms
in a system

1. $H = U + PV$

- a. $H = \text{Enthalpy}$

- b. $P = \text{Pressure}$

- c. $V = \text{Volume}$

- d. Whenever V changes, P changes to respond and vice versa. U will change because of it, but H will not.

- ii. U and H change when:

1. Heat is transferred to/from the system

2. Work is done on/by the system

3. Matter is added or removed

- iii. Fuel heating value is total liberated enthalpy

1. $-\Delta h$

d. Entropy (S)

- i. $\Delta S = \frac{\Delta Q_{\text{rev}}}{T}$

1. $Q_{\text{rev}} = \text{Reversible heat}$

2. $T = \text{Temperature}$

3. Usually a given though

e. Gibbs Free Energy (G)

- i. $\Delta G = \Delta H - T\Delta S$

1. $\Delta G > 0 \rightarrow$ *nonspontaneous, cannot generate work*
 2. $\Delta G < 0 \rightarrow$ *spontaneous, can generate work*
- ii. Max work obtainable is energy liberated
1. $-\Delta g$

f. Electrochemistry

i. Basic

1. Charge (Q) = $I * t$
 - a. I = Current
 - b. t = time
2. Energy = $I * \Delta E * t$
 - a. ΔE = *change in voltage*
3. Power = $I * \Delta E$

ii. Charge

1. Maximum work from charge movement
 - a. $q\Delta E$
 - i. q = charge
2. $q = nF$
 - a. n = number of e^- transferred
 - b. F = Faraday constant
3. $E_{cell} = \frac{-\Delta g}{nF}$
4. $\emptyset = \frac{W}{Q} = \frac{\text{Work to bring charge to electrode}}{\text{the charge}} = \text{Potential (unit: volts)}$
 - a. $\Delta \emptyset = \emptyset_{metal} - \emptyset_{solution}$
 - i. In electrode

6. Fuel Cells

a. Efficiency

i. $\eta = \frac{\Delta G(T)}{\Delta H} = \frac{\text{Gibbs free energy}}{\text{enthalpy}}$

$$b. E_{cell} = \frac{-\Delta g}{nF}$$

7. Power to X

$$a. W_{compression} = nRT \ln \frac{P_2}{P_1}$$

- i. W = Work
- ii. R = Gas Constant 8.314
- iii. T = Temperature
- iv. P₂/P₁ = ratio of pressures

$$b. \text{Mol flow} = \frac{\text{Charge}}{nF}$$

$$c. \frac{\text{Energy in}}{LHV} = \text{Efficiency with respect to LHV}$$

$$d. \frac{\text{Energy in}}{HHV} = \text{Efficiency with respect to HHV}$$

8. Energy Storage

$$a. \text{Power density} = \frac{\text{Power}}{\text{Volume}} \text{ or } \frac{\text{Power}}{\text{mass}}$$

$$b. \text{Energy density} = \frac{\text{Energy}}{\text{Volume}} \text{ or } \frac{\text{Energy}}{\text{mass}}$$

$$c. \text{Capacity Factor (CF)}$$

$$i. \frac{\text{Total annual energy}}{\text{Nameplate power} * 1 \text{ year}} = \frac{\text{Total output}}{\text{Theoretical maximum}}$$

9. Batteries

a. Basic

$$i. \text{Charge (Q)} = I * t$$

$$1. I = \text{Current}$$

$$2. t = \text{time}$$

$$ii. \text{Energy} = I * \Delta E * t$$

$$1. \Delta E = \text{change in voltage}$$

- iii. Power = $I * \Delta E$
- iv. Current = $\frac{\Delta Q}{\Delta t}$
- v. Current density = $\frac{\Delta Q}{A * \Delta t}$
 - 1. A = Area

b. Specific

- i. $I = zF * \frac{\Delta n}{\Delta t} = \text{Current}$
 - 1. z = number of e^- involved
 - 2. n = moles of converted metal
 - 3. F = Faraday's constant
- ii. $n = \frac{Q}{zF}$
 - 1. Q = heat
- iii. Thermal Capacity (Q_t) = znF
 - 1. $n = \frac{\text{mass}}{\text{molar mass}}$
 - 2. So for answer of Q in mAh/gram
 - a. $Q_t = \frac{zF}{3.6 * \text{molar mass}}$
- iv. $E_{\text{cell}} = \frac{-\Delta G}{nF}$
- v. Energy stored in a battery
 - 1. $V = E_{\text{cell}} * I * t$
 - a. I = Current
 - b. t = Time

c. Efficiency

- 1. Evoltage = $\frac{\text{Average voltage during discharge}}{\text{Average voltage during charging}}$
- 2. Ecoulombic = $\frac{1}{CF}$
 - a. CF = Capacity Factor
- 3. Efficiency = Evoltage * Ecoulombic

Units

1. 1 Newton = $1kg * 1 \frac{metre}{second^2}$, unit of force
2. 1 Joule = 1 Newton * 1 metre, unit of energy/work
3. 1 Watt = $\frac{1Joule}{1second}$, unit of power
4. 1 Watt hour (1Wh) = $1 Watt * 3600s = 3600 Joules$
 - a. Usually used the other way around

$$\text{i. } \frac{X \text{ Joules}}{3600} = X \text{ Wh}$$

$$\text{ii. } 1 \text{ Joule} = \frac{1}{3600} \text{ Wh}$$

$$\text{b. } 1 \text{ kWh} = 3600000 \text{ Joules} = 3.6 \text{ megajoules}$$

$$5. 1\text{eV (electronvolt)} = 1.6 * 10^{-19} \text{ Joules}$$

$$6. 1 \text{ Joule} = 1 \text{ Volt} * 1 \text{ Amp}$$

$$7. 1 \text{ Nm}^3 = 1 \text{ Normal cubic meter of hydrogen at 1 atm 0 C}$$

$$8. 1 \text{ Ampere} = \frac{1 \text{ Coulomb}}{\text{second}}$$

$$\text{a. } 1 \text{ Coulomb} = 1 \text{ Ampere} * 1 \text{ second (unit of charge)}$$

$$\text{b. } 1 \text{ Ampere hour (Ah)} = 3600 \text{ Coulombs}$$

$$\text{i. } 1 \text{ mAh} = 3.6 \text{ C}$$

Constants

$$1. E_{\text{cell}} = \frac{-\Delta G}{nF} = 1.23\text{V at 1 bar, 25 C for fuel cells}$$

$$\text{a. The mathematical perfect thermoneutral voltage is 1.48V}$$

$$2. F = \text{Faraday Constant} = 96485 \frac{\text{sA}}{\text{mol}}$$

$$\text{a. s = seconds}$$

$$\text{b. A = current}$$